

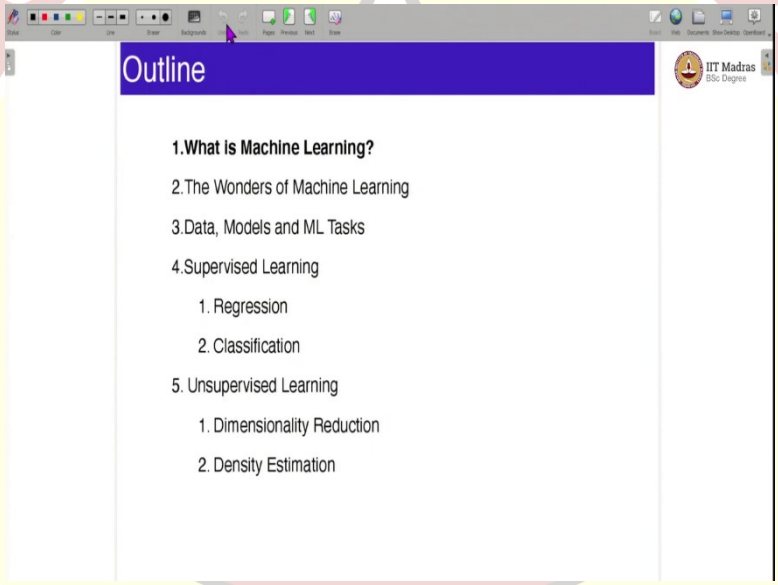
IIT Madras

ONLINE DEGREE

Machine Learning Foundations
Professor Harish Guruprasad Ramaswamy
Assistant Professor, Department of Computer Sciences & Engineering
Indian Institute of Technology Madras
What is Machine Learning?

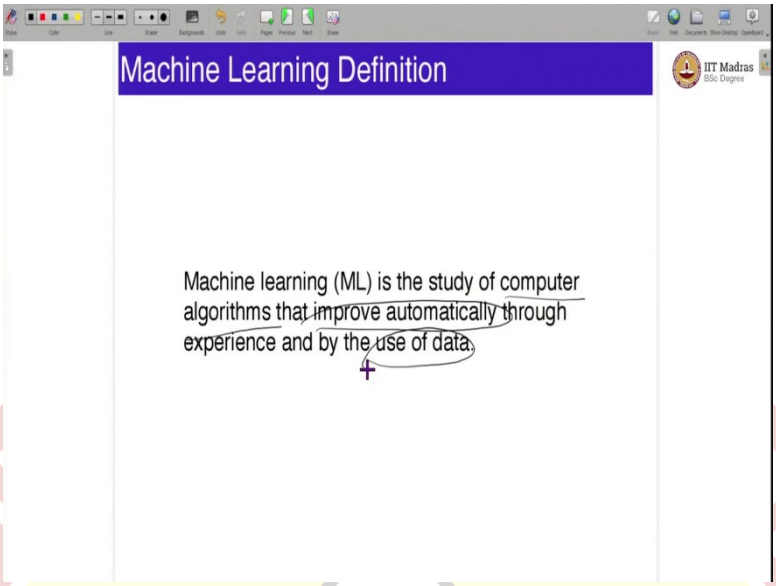
Hello, students, welcome to the course on Machine Learning Foundations. I am Harish Ramaswamy, and I am one of the three instructors who is responsible for teaching this course. The first week, we will take a brief look at the basic foundation, basic terminology and introduction of this course and set up some problems, which will be alluding it throughout the course.

(Refer Slide Time: 00:42)



First, here is a brief outline of the content for this week. The first question that everyone should be asking is, what is machine learning? You must, might have heard of this term, so bandied about in the press in the newspaper, in the TVs, but what is machine learning?

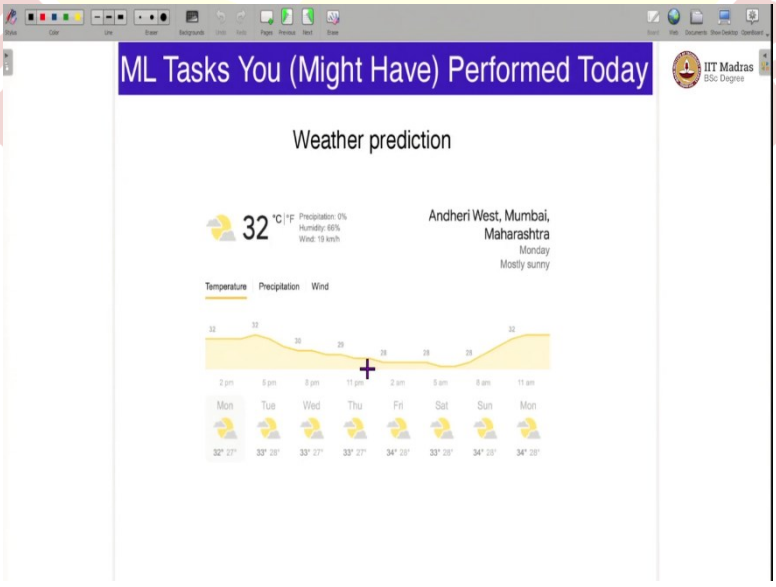
(Refer Slide Time: 00:57)



According to Wikipedia, this is the definition that you can find in Wikipedia. It says machine learning are, which is popularly abbreviated as ML, is a study of computer algorithms that improve automatically through experience and by the use of data. This is a pretty concise definition, and it does a pretty good job, but it uses more terms than can be understood easily.

For example, what is a computer algorithm? And what do we mean by improve automatically? And what exactly is the use of data? These terms are not fully clear, even though they are reasonably approximately clear. So, we will throughout the rest of this course we will make these terms clear.

(Refer Slide Time: 01:41)



Before we even make that attempt to just to impress upon you the importance and the ubiquity of machine learning tasks even before coming and watching this video today, you might have performed several machine learning tasks. I will give two examples. One example might be weather prediction.

You might have just after you woke up you might have just Googled what is the weather now. And Google or some other weather site might have given you an output like this. While machine learning is by no means the only tool that is used for giving weather prediction it is a major reason why weather prediction is so much more successful these days than let us say 20, 30 years ago.

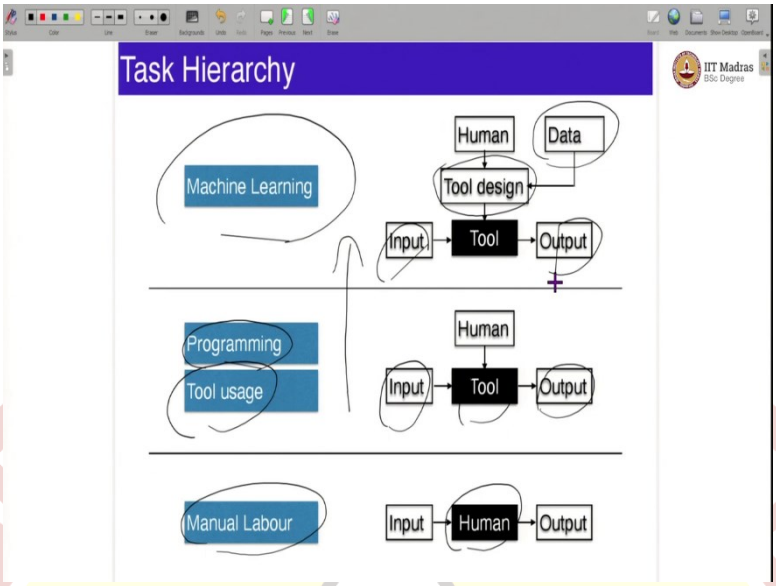
(Refer Slide Time: 02:25)



Here is one other example of a machine learning tasks that you might have performed today even before let us say watching this video. Let us say you wanted to take a group picture of your friends and family. And any reasonably advanced camera software these days would automatically detect the faces in the group of people, and it might put boxes so that it is easy for you to focus, it is easy for the camera to focus.

And autofocus would automatically focus on the faces in the image because most people who take pictures of faces, want the faces to be in focus and not the background. Machine learning is an important and in fact the most important tool that is used for doing this kind of task. I have been saying the word task several times over and over. What do we mean by that?

(Refer Slide Time: 03:18)



Task has an English meaning, as it is, but what when we say task. We mean it in the sense that a task is a process, which converts an input into an output that is the notion of abstraction that we will be working with. A task is something which converts an input to an output. So, a task can be performed at various levels.

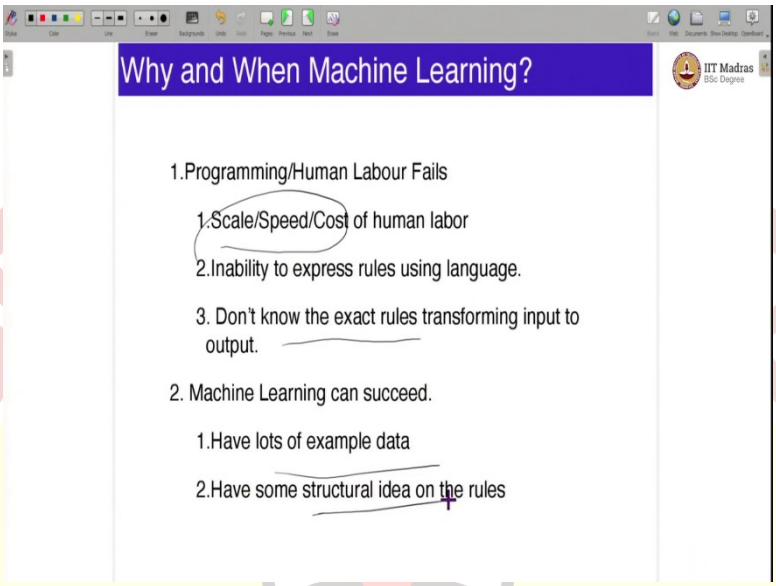
The lowest level is the level of manual labor where a human directly operates on the input and converts it to an output. So, that is the lowest level of that any task can be performed that is a manual labour. One level above that you have tool usage where the human does not directly operate on the input to create output, but rather the human constructs a tool, which in fact, operates on the input and converts into an output.

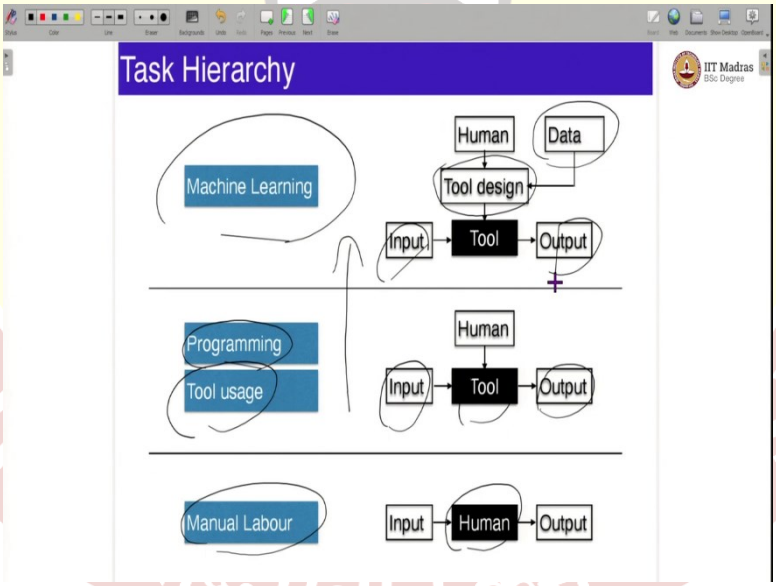
A special case of tool usage is simply programming where the tool happens to be a computer. And the way the human creates a tool is typically through programming and software. Then you have the next level of abstraction, which is machine learning. Here, the human actually does not even build the tool himself. What the human does is, actually just use a broad blueprint to a tool design box, which in fact, constructs the tool.

It also uses an additional source of data along with human inputs to make the tool. So, you can clearly see that the higher up you go the level of abstraction is increasing. At the lowest level the human directly operates on the input and converts it to the output at the highest level you have version where the human actually does not even construct the tool which converts the input into output.

The human just gives broad instructions to a tool design, which in fact uses the data and constructs the tool, which converts the input into an output. All of this might sound a little abstract. Now, we will give several examples and make this clear.

(Refer Slide Time: 05:17)





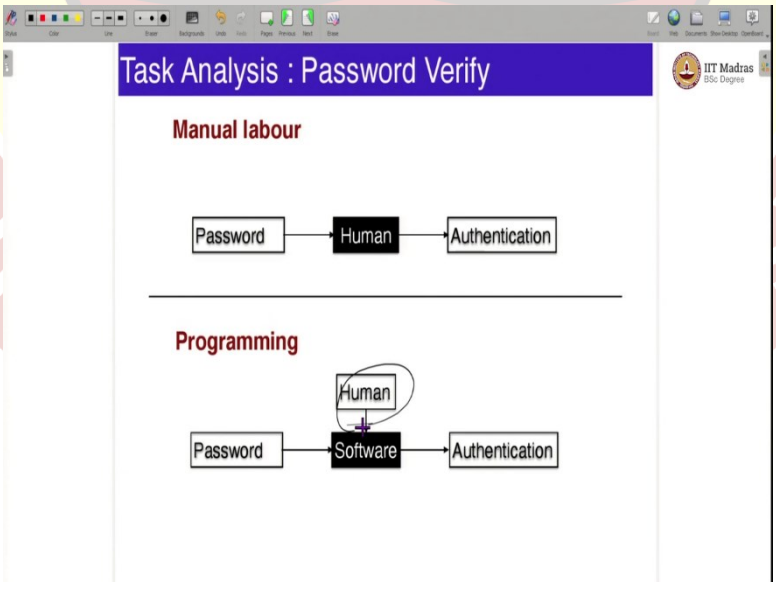
The first question before we do that, you should ask why should you use machine learning at all? Machine learning seems significantly more complex than manual labor or tool usage. Why should we even use machine learning? The answer to that is if you can use programming or human labor to do a task, you should not use machine learning, you should use simple programming or human labor.

But there are several situations where programming or human labor will just fail. The main reason why human labor would fail is because it just does not possible to give, achieve the task at the scale, or the speed or the cost that is available to perform that task. This is the common reason why programming might fail in performing a task is that you might not be able to express the rules that convert the input into an output using say computer programming language.

Or in some cases, you do not even know the exact rules that transform the input into an output. So, these are all reasons why programming or human labor can fail at a task. But that is not sufficient for machine learning to succeed. For machine learning to succeed, we need one other thing, which is we need to have sufficient hope that machine learning can succeed.

And one of the main reasons we think machine learning can succeed is because if it has access to data. If you have access to a lot of example data, then machine learning can reasonably succeed. In addition to that, it is also nice if you have some structural idea on the rule. Even though you do not know the exact rule that transforms the input into an output, you must have some structural idea on the rules. So, all these points will make clear when we look at several examples now.

(Refer Slide Time: 07:05)



Let us consider a very simple task, first. Let us consider this, let us call this task password verify. So, let us say you are Google now, and the task is to verify the password input by some human somewhere. Can this be performed? Can this task be performed at the level of manual labor?

Conceivably yes, that is, when someone types a password, the password gets sent to a server and there is a human who is actually looking at the server and just checks the password and then sees whether the password matches the password on file and then gives authentication. It is in principle possible for a human to do it.

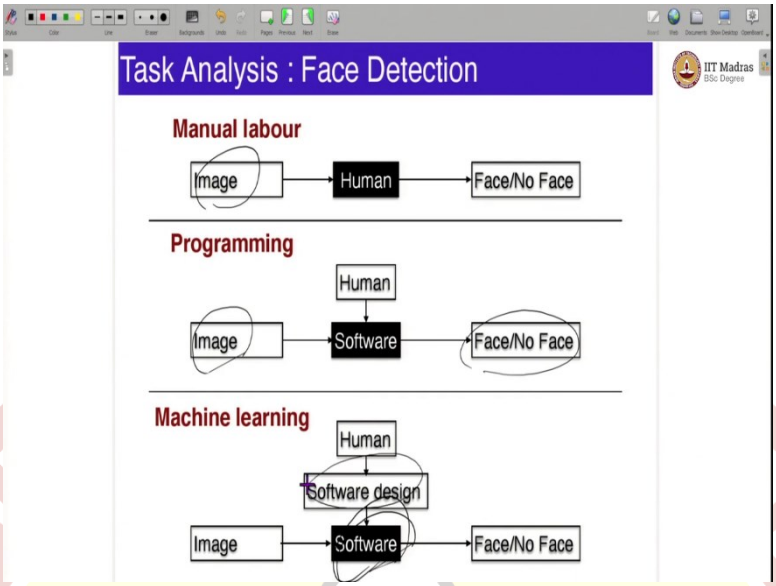
But you can clearly see it is not possible in scale that is we will need billions of humans just to do password verification, which is not clearly possible. And humans are more likely to make errors and so on. Then the, so this particular task is not suitable for to be performed at the level of manual labor. Can this task be performed at the level of say programming?

And in fact, the answer is yes. Because a human does not need to do this task of converting a password input into an authentication output directly. You can just write a piece of software, which in this case is the tool, which just checks whether the password that is input matches the password on file, and if it does, it gives the authentication and that well you can now log into your ID or if it does not match, then it can say well authentication is denied.

This is a particularly simple piece of software that a human can directly design and directly write. There is no need to go to higher level of complexity or abstraction where a human does not even directly write the tool, but because a human can easily write the tool in this case and can easily create the tool for solving the problem of this particular task.

So, this is an example of a task which is not suitable for manual labor, but it can be solved using programming. Because it can be solved using programming, there is no need to even consider going into the level of abstraction of machine learning.

(Refer Slide Time: 09:09)



Now, let us consider another task that we just discussed now, that is the task of face detection. In this case, the input let us imagine the thing the camera itself as the, let us say you are the camera itself, what is the input and what is the output? Well, the camera has to take the image as the input and it has to say whether well this particular image has it a face, does it have a face or not? If it has a face, where is the face?

It should, that is the output of this particular task. Can a human do this? Conceivably, once again, yes, a human can do it, that is, whenever you take your new Canon camera and you focus on a group photo that particular, whenever you press the button, whenever you are about to press the button that particular image is sent on a server to let us say where there is a human looking at all the photos that are about to be clicked, and he would just go and draw boxes on those images.

And those boxes are then somehow conveyed to the camera and then the camera once again displays the box. It is conceivable, it is possible to do it, but it is clearly not feasible or efficient because there are just not enough number of humans to draw boxes on all possible images. So, that rules out manual labor. Can this be performed using programming?

By that we mean that there is a human who writes a piece of software directly onto the camera. And what does this software supposed to do? Well, this software is supposed to take or let us say a programming function which is supposed to take an image as input and it is supposed to output the location of the faces in the image.

Is this possible? I mean, can a human write a program which will do this? Immediately you can see that it is not straightforward to write a function which does this. Why is that? Because while humans have an idea of what constitutes a face, they cannot really write it in a mathematical, mathematically precise way, what aspects of this pixels in an image constitute a face and which does not constitute a face.

You can easily, let us say, if you are teaching the concept of a face to a baby, you should, you would be able to easily do that. You can point to a baby and say this is a face, this is not a face, this is a face and this is not a face, and the baby would pretty quickly understand it that these are faces and these are not faces. And when you give to the baby an image containing several people the baby can easily identify which where the faces are in that image.

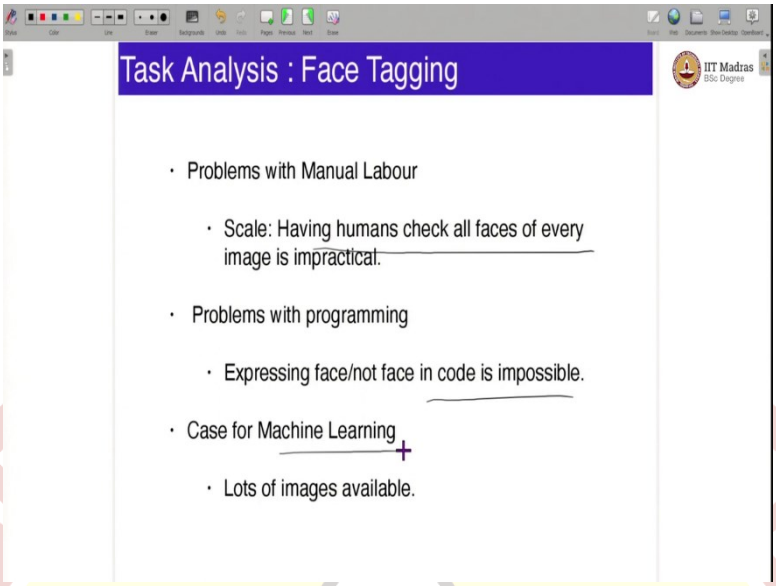
So, that is you know what a face is. It is just that you cannot convey that idea to a computer efficiently because computer does not look at it. As far as the computer is concerned, every image is simply a collection of pixels. The idea of a face cannot be conveyed efficiently to a computer, which just knows pixels, even though there exists, there clearly exists a function which can do that.

There exists a function, which can take in an image which just constant, which is just made up of pixels and say where the faces are, that clearly exists such a function, but just that a human cannot directly write a function because he is not able to convey the idea of a face to a computer directly. So, this rules out programming also.

Now, this is an example case where machine learning can make a huge advantage. Because here, the human does not need to even know, does not need to exactly convey the idea, the exact tool, because the human is not actually building the tool. The human is just giving the broad paradigms of the software design and the software is not directly built by the human.

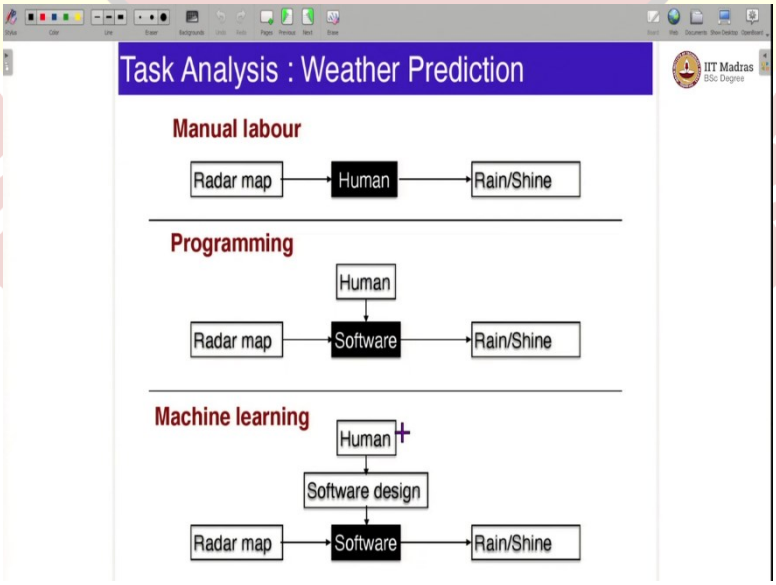
Of course, you need lots of data. And this is one example where face data is, face image data is practically unlimited on the internet, so there is no dearth of face data. So, you have a lot of hope of success with machine learning because there is lots of face data and humans do not need to exactly they do not have to build the tool from scratch because they do not have to write the software themselves, they only have to give inputs to a software design, which is what creates the software that converts an image into a face.

(Refer Slide Time: 13:30)



So, here is a summary of the problems of the face tagging task. That is the problem with manual labor is that having humans check all phases of every image is just impractical. And the problem with programming is that expressing face or not face in code is just impossible. The case for machine learning is that you do not have to express this face not face in code, as a function of pixels directly you just have to give lots of images, which are faces and lots of images which are not faces. This is sufficient for a machine learning paradigm to work.

(Refer Slide Time: 14:10)



Here is one other final task that we will be analyzing, which is considered the task of weather prediction. So, weather prediction, what is the input and what is the output of the task. The

input of the task is, a radar and map. Let us say, you want to predict the weather or let us say, you want to predict whether Mumbai is going to get rain or not the next day, that is your goal. And what is the input for making such a prediction.

Let us say you have the radar map of let us say the entire Arabian Sea, Bay of Bengal and India and so on. So, you have the radar map of containing the information about the wind, about the moisture and all that, you have that as the input for you and you want to predict whether tomorrow Mumbai is going to get rain or it is going to be sunny, that is the goal.

And in fact, in this case, it is not even possible for humans to do it because humans do not know, given the radar map, whether it is going to rain or not. Because there clearly exists such a function because physics says that there exists because whether it is going to rain or not tomorrow clearly depends on the moisture, the wind and other conditions that are prevalent today.

So, there clearly exists a function, which is what is implemented by nature when it decides to rain or not the next day, but just that humans do not know this function. So, in fact, in this particular case, the human cannot actually, this particular problem is not able to be solved using manual labor, because the human does not know the function which maps the input to the output.

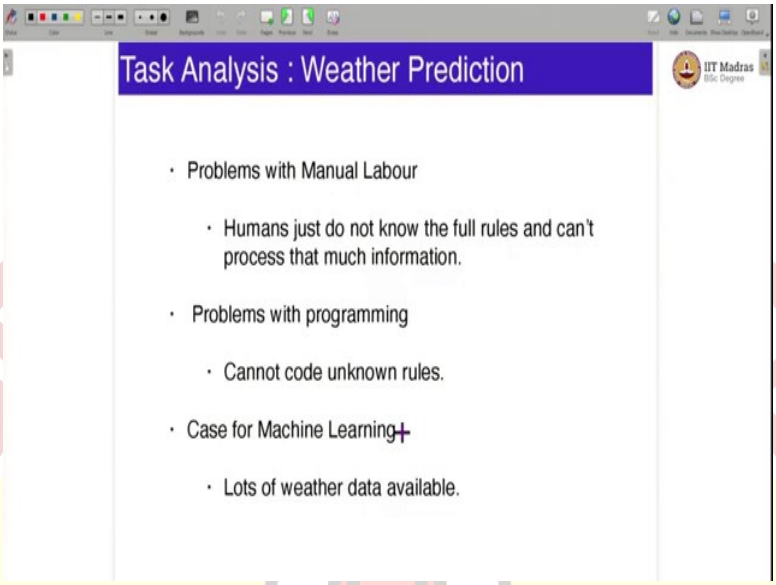
It is for that exact same reason programming will also fail, because if the human does not know what transforms the input to the output there is no way he or she can convey that idea to a computer. So, programming, this particular task cannot be solved with the level of programming as well.

At the level of machine learning, though, immediately it becomes possible because the only requirement is that there exists a rule which transforms the input, which is in this case the radar map to whether it is going to rain or not the next day, and we know such function exists because nature itself is implementing such a function. You do not have to know that function, you do not have to be able to convey that function to a computer and so on.

So, the only thing you need to know is that, there exists such a function. And in fact, you know that exists, so you do not have to worry about the problems that you encounter with manual labor and programming. Once again, the human just has to give the broad parameters of the software design, which in turn designs a software and uses some additional data of weather

maps and creates a software which will convert the let us say the local radar map into whether tomorrow we are going to get rain or shine.

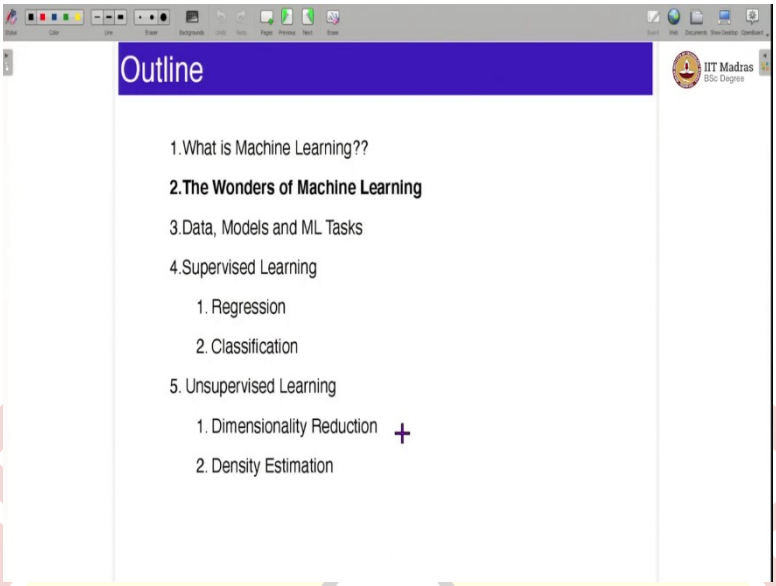
(Refer Slide Time: 17:09)



So, this is another example of a case where machine learning is an example of a good machine learning task, that is, you cannot solve it with manual labor and you cannot solve it with programming also, but you can solve it in machine learning because there is hope for solving it with machine learning, because there is lots of weather data available.

And in fact, most weather prediction systems do use machine learning in some way, because of the immense amount of availability of weather data and it is just not possible to predict weather using either manual labor or with just simple programming. So, with that, I think you have a broad idea of what is machine learning and when you can apply machine learning.

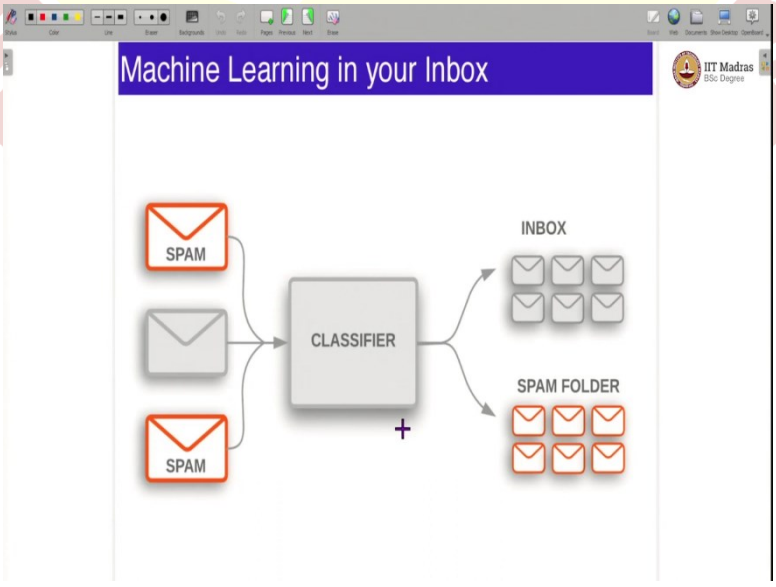
(Refer Slide Time: 17:56)



Then the question becomes, where is machine learning being used? This section is mostly an advertisement of machine learning. Machine learning really does not require an advertisement considering the ubiquitous nature of it, and you just cannot go even a few minutes without encountering machine learning, but I will still go ahead and do that to let you know where all machine learning can be formed.

This is by no means an exhaustive list. There are several applications. Any application that you can think of has a potential use case for machine learning. So, this is by no means an exhaustive list, but this is a couple of examples that you can find machine learning.

(Refer Slide Time: 18:35)



Let us get the easy example out of the way first, which is an example that you encounter every day in your email inbox. Any reasonable inbox system like Gmail or Hotmail or any reasonable email system actually has a very good spam classifier in it, that is, every time you receive an email the mailbox itself classifies that email as either spam or a regular email.

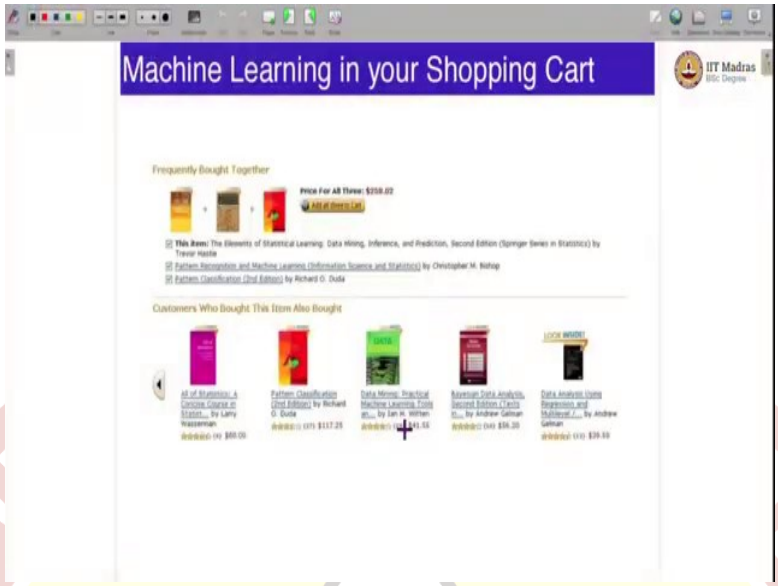
And all the spam emails get into a spam box, and all the regular emails just go to the inbox. In fact, this is one of the biggest success cases of machine learning where machine learning has largely solved the problem of spam. Even let us say 10 years ago, spam was a huge problem, that is, for every valid email that you get there would be 5 to 10 spam emails, which you would have to sift through to get to your regular email, but machine learning has tamed the problem of spam and it has successfully done so.

And why was this problem not? We can ask the question this is a machine learning task. Why cannot it be solved using human labor? Yes, it can be solved using human labor, but the problem with human labor is that it has the problems of scale and speed. The problem with programming is once again similar to the problem with the face detection, where, given an email how do you say with the program that this particular email is spam or not spam.

Because spam has the righty of characteristics, you cannot do any simple function which, just write a function which takes in an email and says whether it is a spam or not. There is no simple function, which can do that. Given an email, you can say whether it is mostly you can say whether it is spam or not, but it is just not possible to convey this idea to a computer.

But once again, this is easily solved using machine learning, because you do not need to convey that idea at all. The only thing you need to give to the system is lots of data, which in this case boils down to giving it examples of lots of spam emails, and examples of lots of non-spam emails, that is sufficient for the machine learning algorithm to design the tool that actually takes in these inputs, which are emails and classify them as spam or not spam.

(Refer Slide Time: 20:59)



So, and one other example that you encounter every day is in your shopping cart. Where you go to Amazon, and you say you buy a couple of books, so you will often see this extra thing, which will say customer who bought this item also bought these other books. This is an example of what is called a recommender system and recommender systems are another important task that is often solved using machine learning.

In fact, when you go to YouTube or when you go to Netflix, this is again, brought to it fore when you watch a TV show then you get a bunch of other TV shows that says people who watch this TV show who want who like this TV show also like these TV shows. And similarly for YouTube, when you watch one particular news video, it will automatically recommend other videos which other people similar to you have watched.

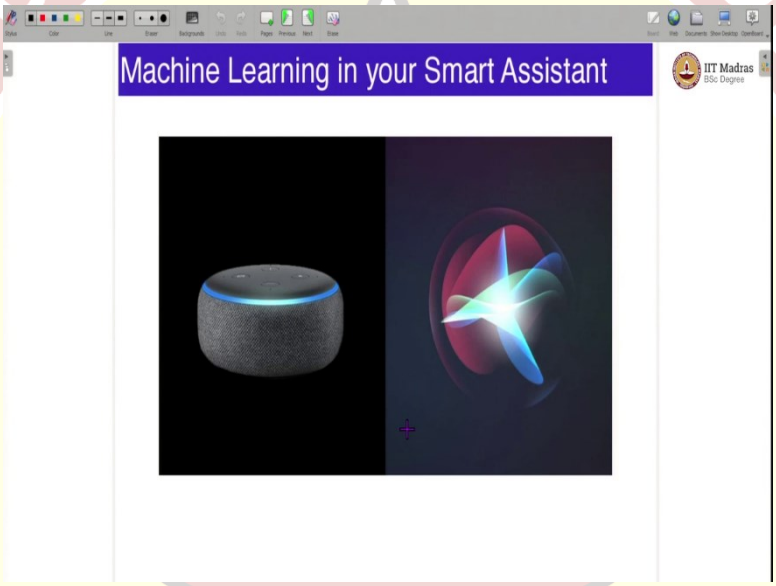
So, this is how does machine learning do that we will not be attacking that particular problem in this course, but that is an example of machine learning. Once again, can this be solved using humans? Potentially, yes, that is a human can just do a psychological analysis of each person, and then say, okay, this person likes these videos, so this is a type A kind of person, so he will like these kinds of videos also.

But you can clearly see this is not scalable to all humans, and it seems like a wild violation of privacy. And this cannot be done using programming because you might; there is no clear rules on which video someone else is going to like given that he has watched these other videos. Even though you might have a hunch, it is difficult to translate this hunch into at the level of programming where you can convey that to a computer directly.

But to solve this task at the machine learning level, you need neither. You just need to have examples of several examples of videos that this person has watched and other people have watched, and we just use that to come up with a tool which takes in the input that is your history of videos that you have watched our books that you have purchased and predict future videos that you will like are future books that you will like.

So, this is an input, output, which is not possible efficiently with humans and not at all possible with program and with the level of programming, but it is possible to solve this particular problem also at the level of machine learning.

(Refer Slide Time: 23:22)



Another example, that you might have encountered very often is see your smart assistant is your Alexa, is your Siri, is your Google Assistant and so, on all of these use machine learning very, very integral part. Right from the part that when you speak, your sound waveforms are converted into text and which in turn is converted into commands.

As far as the computer is concerned, a sound waveform is just a waveform. It is a bunch; it is a time varying signal, that is it. It is not, I mean, it is possible to solve this at the level of humans where you hide a small human inside every smart speaker and that small human decides whether what you are saying is Alexa or not.

But you can clearly see the ridiculousness of it, but you cannot, this solution is simply not scalable. But you so, well, can you solve this problem using programming. Once again, you run into the same problem that you had with face tagging, with spam detection and so on. While

you might be able to take here a piece of here a sound and say whether that sound was Alexa or not Alexa. How do you do that at the level of programming?

How do you write the program? How do you write a function, which takes in a time varying signal as input and output whether the word Alexa was said or not said in that particular signal? It is immediately the problem is that you cannot convey the idea of the sound Alexa to a computer efficiently.

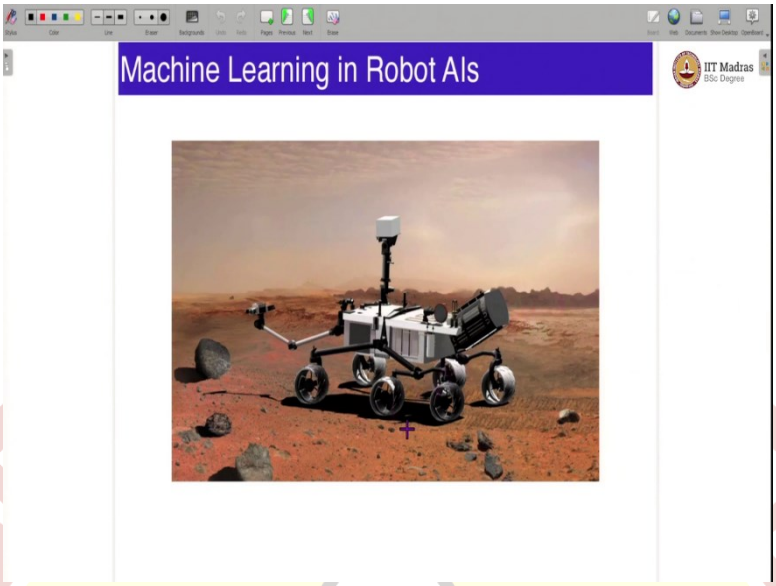
Once again, machine learning has no such issues with doing that because you only need to give several examples of Alexa and several examples of not Alexa, and the machine learning tool design will automatically do the job for you and it will convert the, it will construct a tool, which will convert your input audio into whether the sound was Alexa or not Alexa.

And of course, smart assistants use ML even more than this that is it, they do not just convert sound to text. So, they also convert text into commands that can be actually performed by the smartest. Let us say if you said 'increase volume by 10 percent'.

Let us say that is what you said, it has to first convert what you say that is increase the waveform corresponding to increase volume by 10 percent, into the words increase volume by 10 percent and then it has to detect that the command that you have given is corresponds to increasing volume by 10 percent, because not everyone might say it in the same way.

They might say things like, raise sound by one point. They might say, which might pretty much mean the same thing, colloquially, but how do you convert all of that into a reasonable command that an assistant can perform on the mobile or on any system. So, machine learning is used for that as well, machine learning is used for converting the text command, a text into a command that can be performed by the assistant.

(Refer Slide Time: 26:15)



And one other example, that used to be all the rage that used to be one of the main motivations behind developing machine learning itself was building a robot artificial intelligences. If you have seen any movie like Terminator or Robocop or iRobot or anything like that, you know that these used to be all the rage in pop culture.

And they are still one of the main goals of machine learning, but learning has formed so many other successes that it is not the only goal anymore. But it is still useful in several robots. For example, when you build, let us say, in this case, this is the I think the perseverance rover in Mars.

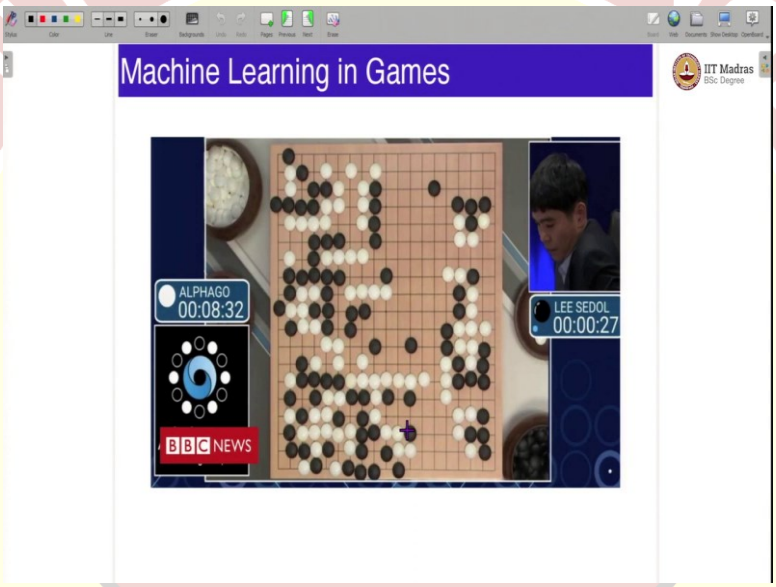
So, when you build a robot for, let us say, operating in Mars, it is just not possible for humans to do the job simply because it is going to take 10 to 15 minutes for just the radio signals from Mars to reach Earth. So, you cannot remote control the robot using, it is just not possible. So, there must be, you can do some level of programming, you can perhaps pre-build a route of let us say you want the rover to move from crater A to crater B.

You might have already computed the route and said, broadly, this is the thing that you should follow. But there are several unexpected things that can happen, and a programming just cannot handle it. So, for example, in this case, an input situation, input might be some environmental situation, and output might be what action to take.

The action might be, let us say, move forward, more left, stay there, switch on your parachute or huddle up or there might be several actions that a robot can do that might be the output and the input would be the current environment.

Once again, it is not possible to hard code program, hard code a program for all possible variations simply because you do not know what is going to happen, but machine learning can handle this to a certain extent, because you just need to give enough data to it so that it can effectively find the right or a good thing to do for every possible situation.

(Refer Slide Time: 28:26)



And one other recent huge success on machine learning has been in games. We know that in chess already in the late 90s there was human built chess system which beat the best humans, best human, just Garry Kasparov. But these systems have had even more success in recent times where for example, GO, was supposed to be an impossibly complex game that computers were not supposed to make for the next 10, 20 years.

But suddenly, immediately the, this thing happened in the quite recently in just few years ago where a GO playing system which was primarily built around machine learning beat the best human GO play. And this is eventually going to happen for pretty much all games with complete information. Artificial Intelligence or machine learning agents are the most powerful players.

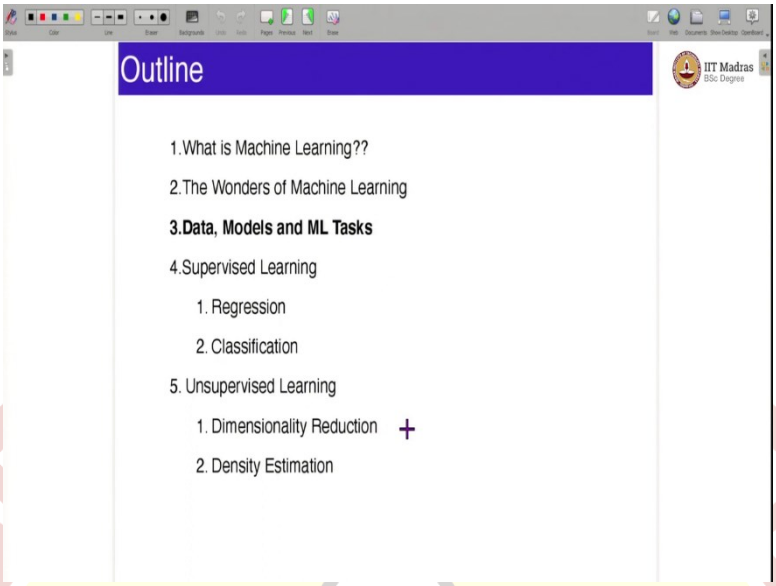
(Refer Slide Time: 29:28)



And in practical industry applications, machine learning is used in for example, in things like marketing. In social media marketing where you might have a limited budget. Let us say you have a budget of 100 dollars and Facebook is asking 10 dollars for a single ad, you might that means that you have a budget for 10 ads. But whom do you advertise to?

So, this is another, if you are a brand you have limited budget, but you want to reach as many people as possible, how do you decide whom to advertise to so that you can maximize your impact? So, this, how do you market across, let us say some social networks, how do you market across Twitter or YouTube or how do you do this efficiently is a very complicated question, and machine learning has made huge strides in solving such problems.

(Refer Slide Time: 30:20)



So, with that, I think this helps you to understand what is machine learning and when can you apply machine learning. And finally, you also know several areas where machine learning has been applied. So, this gives a brief introduction into machine learning. We will jump into the nitty gritty details of what makes machine learning work in the next section. So, for this, that is all we have for this video. Thanks.